

Executable Self-Justifying Axiom Systems over U-Grounding Arithmetic: Mechanizing Willard’s Tableau Apparatus in a Weak-Arithmetic Setting

Willard’s self-justifying axiom systems (SJAS) form a family of weak arithmetical theories that can consistently extend themselves with a carefully formulated self-consistency principle, despite living in the shadow of Gödel’s second incompleteness theorem. The construction is not merely a model-theoretic curiosity: it depends on a precise interaction between (i) a *U-grounding* arithmetic base, in which addition is total but multiplication is available only as a graph relation $\text{mult}(x, y, z)$ rather than a function symbol, and (ii) a reflected semantic-tableau proof apparatus D , together with arithmetized predicates for codes of formulae, proofs, and finite axiom bases β . In an $IS\#_D(\beta)$ system, the self-consistency sentence asserts that no D -proof derives the code of a contradiction from the code of the finite system β . The boundary at which self-justification becomes possible is therefore proof-theoretic and intensional: it is the failure of a particular encoded tableau to close under the rules and size constraints Willard imposes, not an extensional identification of provability with truth in a stronger theory.

This talk reports on an executable mechanization of the $IS\#_D(\beta)$ line in *Proflog*, a logic programming language whose operational semantics is Fitting/Smullyan semantic tableaux with equality and subsidiary procedure calls. The implementation makes the weak-arithmetic object language, the reflected clause basis, and the arithmetized proof predicate $\text{TabPrf}_\beta(s, t, p)$ —“there is a semantic-tableau proof p of theorem t in system s ”—first-class objects that can be queried, executed, and audited at the proof-certificate level. Because SJAS self-reference is sensitive to proof-tree branching, closure rules, and encoded proof size (including Willard’s conventional tableaux encoding requirement), the mechanization is organized around *visible* tableau descent from source formulae to kernel proof terms, rather than host-language evaluation of truth values.

From the perspective of weak arithmetics, three aspects of the work may be of particular interest. First, the U-grounding design is treated as an executable signature and coding discipline, not only as a metatheoretic hypothesis: numerals, bounded formula classes $(\Delta_0^*, \Pi_1^*, \Sigma_1^*)$, and Level-1 self-consistency schemata are generated and classified inside the same first-order setting used for ordinary programs. Second, the proof predicate is implemented as a structural checker over encoded tableau trees, aligning with Willard’s semantic-tableau proof predicate for ordinary tableaux; ongoing correspondence work establishes a biconditional over a sharply bounded first fragment of certificates, with equality, procedure-call, and quantifier constructors shown unreachable in that fragment. Third, the executable setting supports *computational* exploration of Willard’s “strength dial”: the same tableau kernel runs Fitting-style reference programs and SJAS-encoded proof obligations, showing where tableaux close, remain open, or become unresolved when coding strength changes.

The talk will summarize the weak-arithmetic commitments of the mechanized $IS\#_D(\beta)$ builder, exhibit arithmetized proof checking on small encoded examples, and state honestly what is and is not yet proved: full Willard equivalence, U-grounding numeral decoding in the correspondence proof, and discharge of β -validity remain explicit trust boundaries. The broader question—which features of a weak-arithmetic language and tableau apparatus must be preserved for self-justification to remain meaningful when the system is made computational—is posed for discussion with the JAF community.

Keywords: weak arithmetic, U-grounding, self-justifying axiom systems, semantic tableaux, proof coding, incompleteness.

References

- [1] D. Willard, Self-verifying axiom systems, the incompleteness theorem and related reflection principles, *J. Symbolic Logic* **66** (2001), 536–596.
- [2] D. Willard, The significance of Π_1 -definable cutsets for incompleteness, feasibility, and self-justification, *Ann. Pure Appl. Logic* **164** (2013), 902–912.
- [3] D. Willard, Addition, multiplication, and self-inspectable theories, *Ann. Pure Appl. Logic* **130** (2004), 27–41.
- [4] M. Fitting, *First-Order Logic and Automated Theorem Proving*, Springer, 1990/1996.